• $P(t) = e^{At}$, Diagonalizar $A = MDM^{-1}$ $\Rightarrow$

$e^{At} = M e^{Dt} M^{-1}$

$$A = \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix}$$

• Eignanvalores : $\det(A - \lambda I) = 0$

$$\det \begin{pmatrix} -1-\lambda & 1 \\ 2 & -2-\lambda \end{pmatrix} = 0 \qquad (-1-\lambda)(-2-\lambda) - (1)(2)$$

$$= (\lambda^2 + 3\lambda + 2) - 2 = 0$$

$$= \lambda(\lambda + 3) = 0$$

$\lambda_1 = 0$ y $\lambda_2 = -3$

• Eigenvectores (vectores propios)

Para $\lambda_1 = 0$

$$\begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow -x + y = 0 \Rightarrow \underline{x = y}$$

$$V_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Para $\lambda_2 = -3$

$$\begin{pmatrix} -1+3 & 1 \\ 2 & -2+3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$2x + y = 0 \Rightarrow y = -2x$

$$V_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

La matriz M

$$M = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix} \qquad D = \begin{pmatrix} 0 & 0 \\ 0 & -3 \end{pmatrix}$$

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$$M^{-1} = -\frac{1}{3}\begin{pmatrix} -2 & -1 \\ -1 & 1 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$$

Exponencial : $e^{At} = M e^{Dt} M^{-1}$

$$e^{Dt} = \begin{pmatrix} e^{0t} & 0 \\ 0 & e^{-3t} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & e^{-3t} \end{pmatrix}$$

$P(t) = M \cdot e^{Dt} \cdot M^{-1}$

$\hookrightarrow M \cdot e^{Dt}$.

$$\begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & e^{-3t} \end{pmatrix} = \begin{pmatrix} 1 & e^{-3t} \\ 1 & -2e^{-3t} \end{pmatrix}$$

$$P(t) = \begin{pmatrix} 1 & e^{-3t} \\ 1 & -2e^{-3t} \end{pmatrix} \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$$

$$P(t) = \frac{1}{3}\begin{pmatrix} 2 + e^{-3t} & 1 - e^{-3t} \\ 2 - 2e^{-3t} & 1 + 2e^{-3t} \end{pmatrix}$$